Chapter 2

LIMITS

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LIMITS

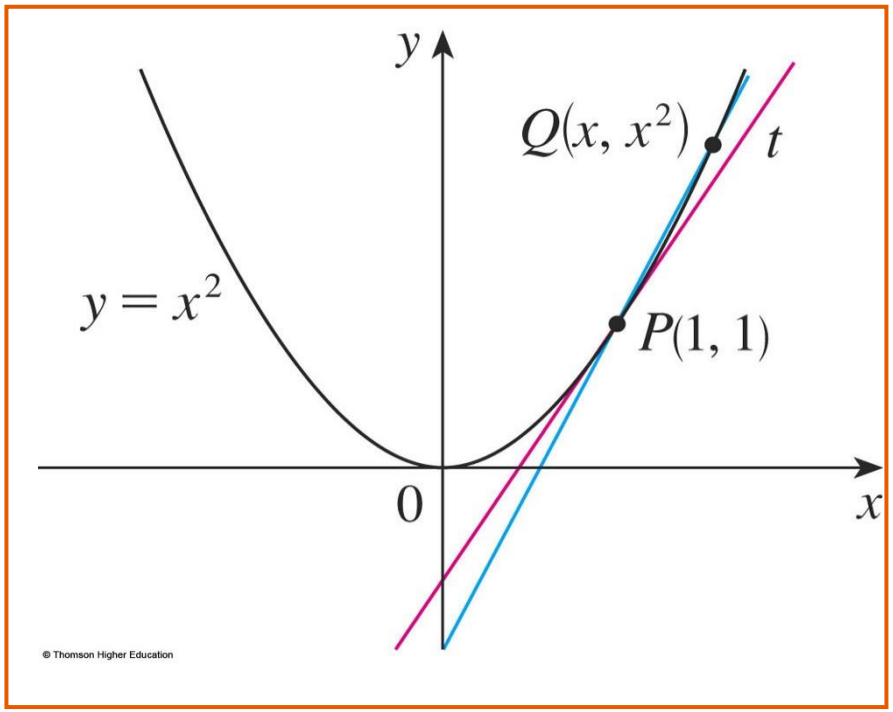
2.1 A Preview of Calculus

THE TANGENT PROBLEM

Example 1

We know that the slope of the secant line PQ is $m_{PQ} = \dfrac{x^2 - 1}{x - 1}$.

How to find an equation of the tangent line to the parabola $y = x^2$ at the point $P(1,1)$?



THE VELOCITY PROBLEM

Example 2

Investigate the example of a falling ball.

- Suppose that a ball is dropped from the upper observation deck of the CN Tower in Toronto, 450 m above the ground.
- Find the velocity of the ball after 5 seconds.



THE VELOCITY PROBLEM

If the distance fallen after t seconds is denoted by $s(t)$ and measured in meters, then Galileo's law is expressed by the following equation.

$$s(t) = 4.9t^2$$

THE VELOCITY PROBLEM

$$\text{average velocity} = \frac{\text{change in position}}{\text{time elapsed}}$$

$$= \frac{s(5.1) - s(5)}{0.1} = 49.49 \text{ m/s}$$

Thus, the (instantaneous) velocity after 5s is:

$v = 49$ m/s

Time interval	Average velocity (m/s)
$5 \leqslant t \leqslant 6$	53.9
$5 \leqslant t \leqslant 5.1$	49.49
$5 \leqslant t \leqslant 5.05$	49.245
$5 \leqslant t \leqslant 5.01$	49.049
$5 \leqslant t \leqslant 5.001$	49.0049

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THE AREA PROBLEM

Example 3

We begin by attempting to solve the area problem:
Find the area of the region S that lies under the curve $y = f(x)$ from a to b.

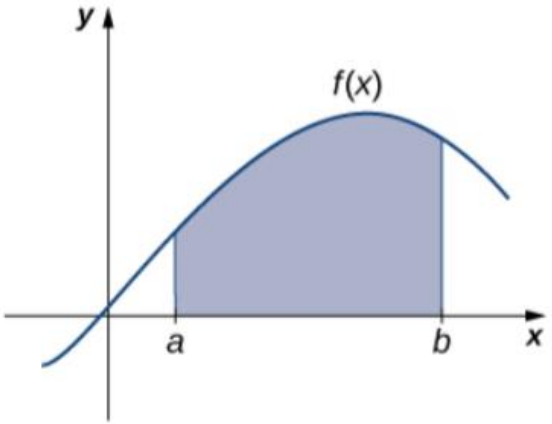


Figure 2.8 The Area Problem: How do we find the area of the shaded region?

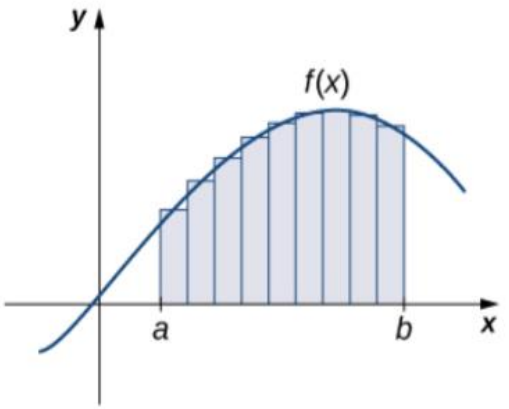


Figure 2.9 The area of the region under the curve is approximated by summing the areas of thin rectangles.

LIMITS

2.2

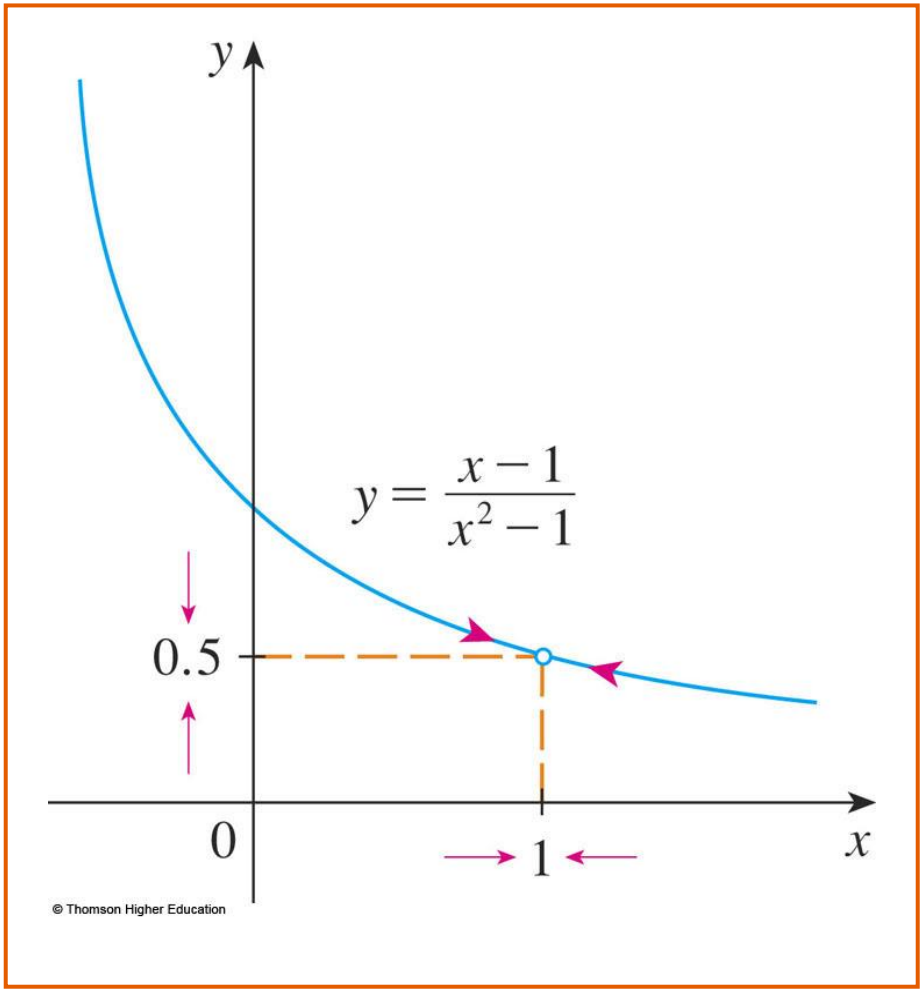
The Limit of a Function

THE LIMIT OF A FUNCTION

In general, we write

$$\lim_{x \to a} f(x) = L$$

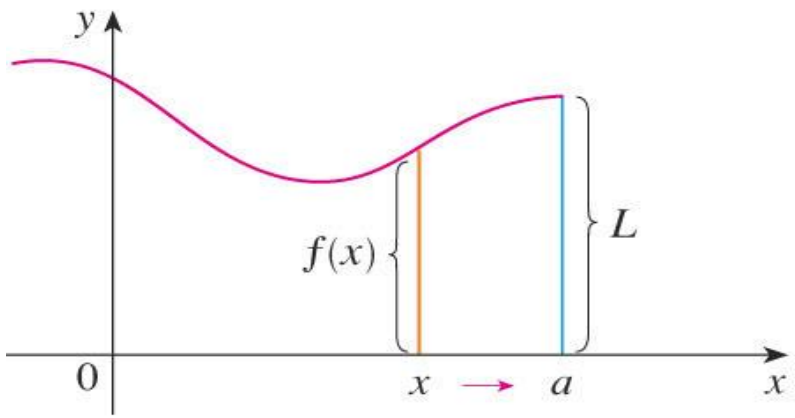
if we can make the values of $f(x)$ arbitrarily close to L by taking x to be sufficiently close to a **but not equal to a.**



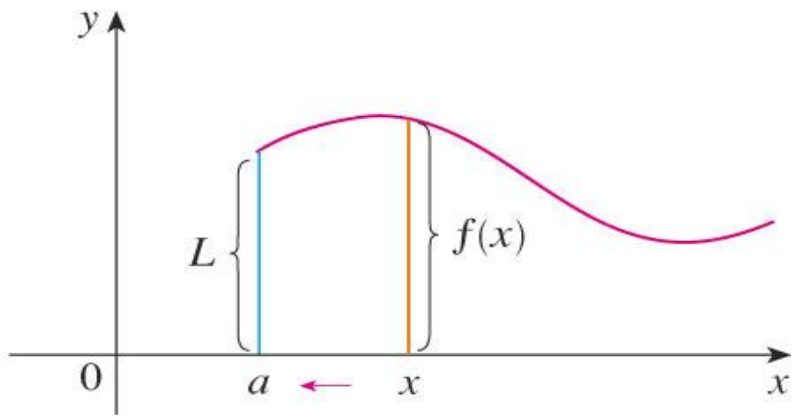
ONE-SIDED LIMITS

We write $\lim_{x \to a^-} f(x) = L$

if we can make the values of $f(x)$ arbitrarily close to L by taking x to be sufficiently close to a and x less than a.



(a) $\lim_{x \to a^-} f(x) = L$



(b) $\lim_{x \to a^+} f(x) = L$

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ONE-SIDED LIMITS

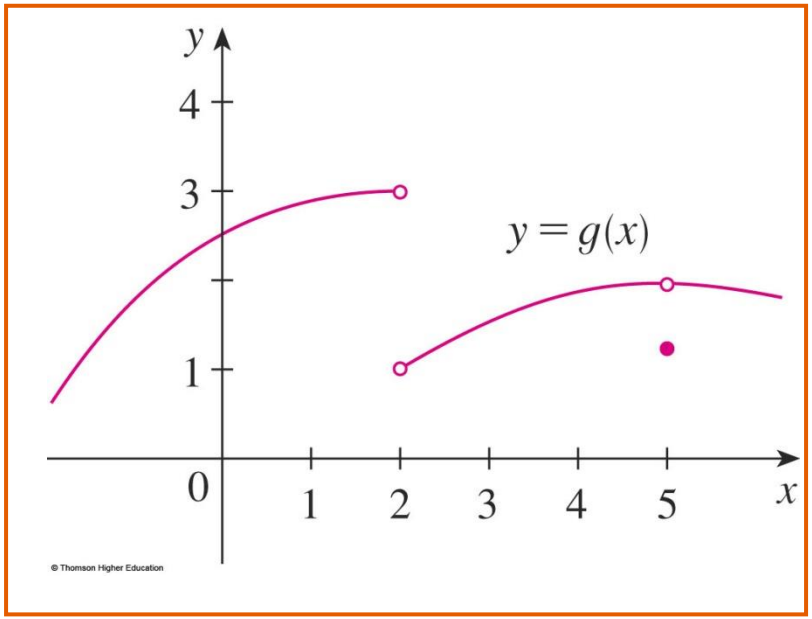
Similarly, *"the right-hand limit of f(x) as x approaches a is equal to L"* and we write

$$\lim_{x \to a^+} f(x) = L$$

$\lim_{x \to 2^-} g(x)$ $\quad$ $\lim_{x \to 2^+} g(x)$

$\lim_{x \to 2} g(x)$ $\quad$ $\lim_{x \to 5^-} g(x)$

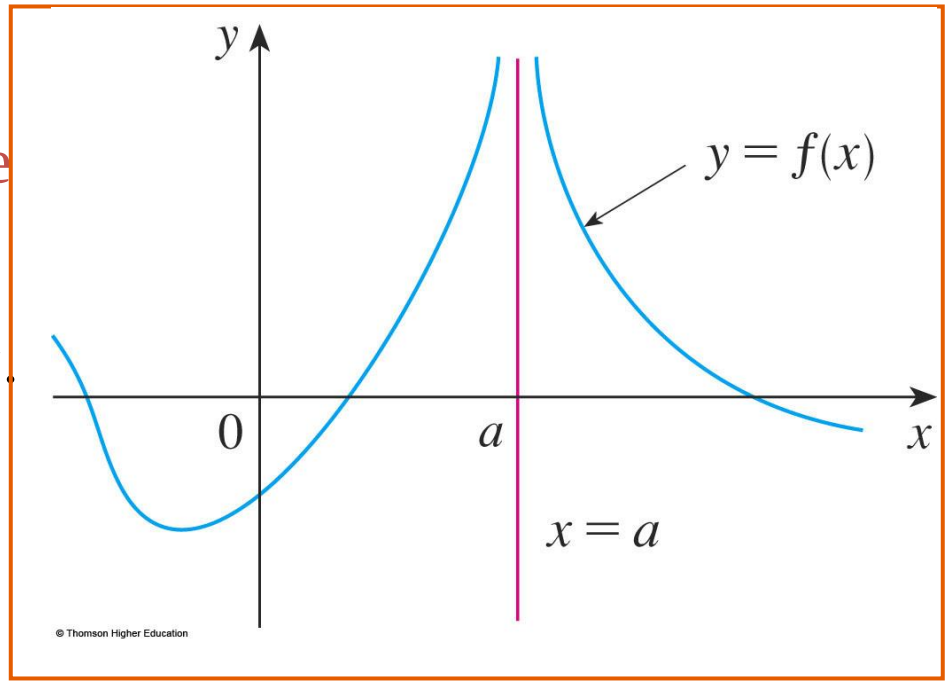
$\lim_{x \to 5^+} g(x)$ $\quad$ $\lim_{x \to 5} g(x)$



INFINITE LIMITS

Let f be a function defined on both sides of a, except possibly at a itself. Then, $\lim_{x \to a} f(x) = \infty$

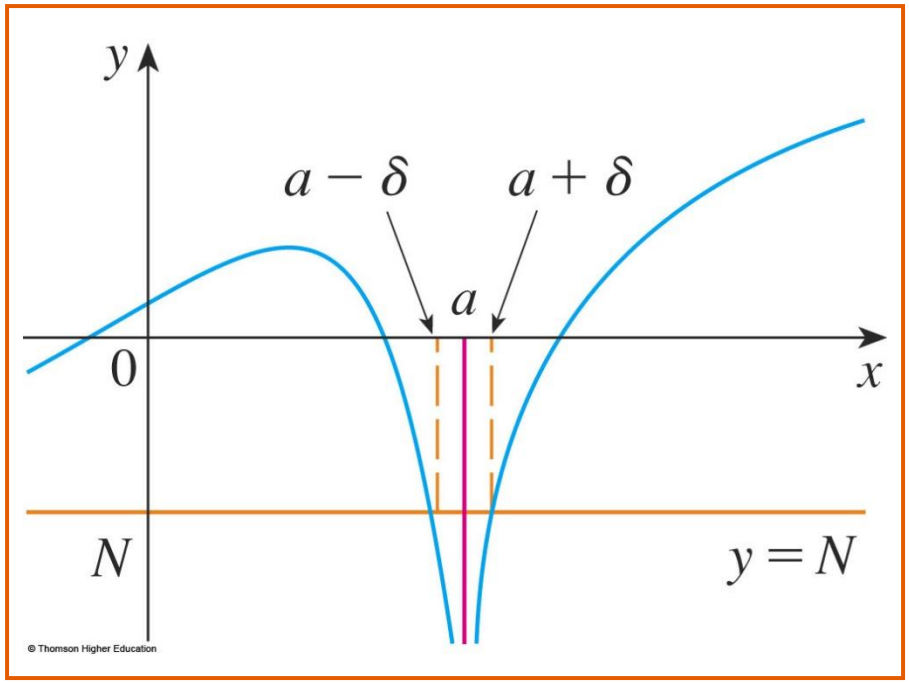
means that the values of $f(x)$ can be made arbitrarily large by taking x sufficiently close to a, *but* not equal to a.



INFINITE LIMITS

Let f be defined on both sides of a, except possibly at a itself. Then, $\lim_{x \to a} f(x) = -\infty$

means that the values of $f(x)$ can be made arbitrarily large negative by taking x sufficiently close to a, but not equal to a.

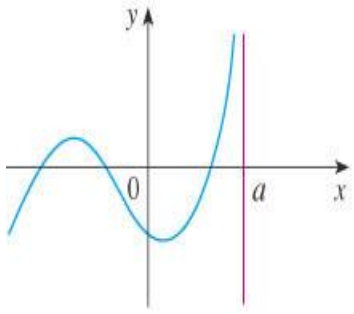


INFINITE LIMITS

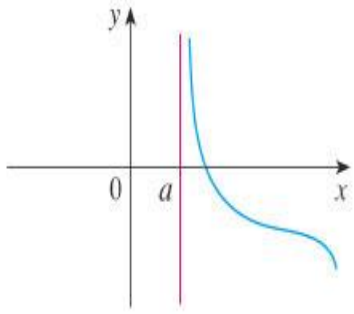
Similar definitions can be given for the one-sided limits:

$$\lim_{x \to a^-} f(x) = \infty \qquad \lim_{x \to a^+} f(x) = \infty$$

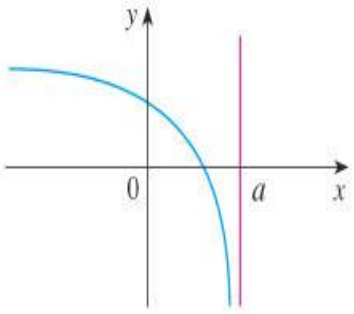
$$\lim_{x \to a^-} f(x) = -\infty \qquad \lim_{x \to a^+} f(x) = -\infty$$



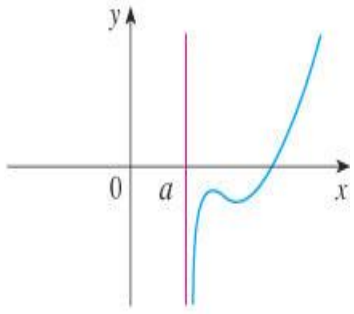
(a) $\lim_{x \to a^-} f(x) = \infty$



(b) $\lim_{x \to a^+} f(x) = \infty$



(c) $\lim_{x \to a^-} f(x) = -\infty$



(d) $\lim_{x \to a^+} f(x) = -\infty$

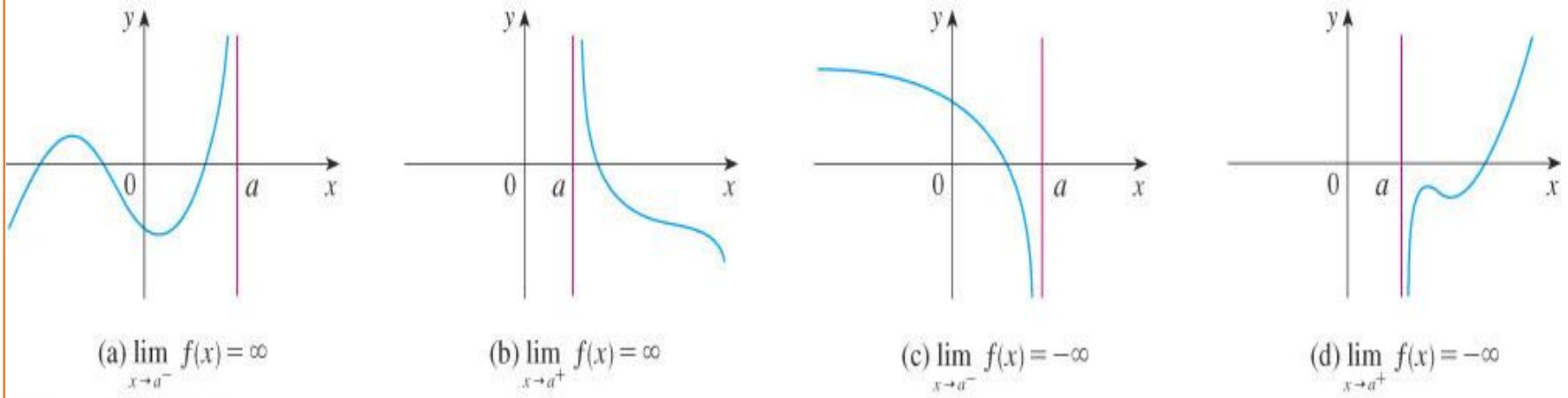
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DEFINITIONS

- x=a is called the vertical asymptote of f(x) if we have one of the following:

$$\lim_{x \to a^-} f(x) = \infty \qquad \lim_{x \to a^+} f(x) = \infty$$

$$\lim_{x \to a^-} f(x) = -\infty \qquad \lim_{x \to a^+} f(x) = -\infty$$



(a) $\lim_{x \to a^-} f(x) = \infty$ (b) $\lim_{x \to a^+} f(x) = \infty$ (c) $\lim_{x \to a^-} f(x) = -\infty$ (d) $\lim_{x \to a^+} f(x) = -\infty$

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LIMITS

2.3 The Limit Laws

In this section, we will:

Use the Limit Laws to calculate limits.

THE LIMIT LAWS

Suppose that c is a constant and the limits $\lim_{x\to a} f(x)$ and $\lim_{x\to a} g(x)$ exist. Then

1. $\lim_{x\to a}\left[f(x) \pm g(x)\right] = \lim_{x\to a} f(x) \pm \lim_{x\to a} g(x)$

2. $\lim_{x\to a}\left[cf(x)\right] = c\lim_{x\to a} f(x)$

3. $\lim_{x\to a}\left[f(x)g(x)\right] = \lim_{x\to a} f(x)\cdot\lim_{x\to a} g(x)$

4. $\lim_{x\to a}\dfrac{f(x)}{g(x)} = \dfrac{\lim_{x\to a} f(x)}{\lim_{x\to a} g(x)}$ if $\lim_{x\to a} g(x) \neq 0$

USING THE LIMIT LAWS

6. $\lim_{x \to a} [f(x)]^n = \left[\lim_{x \to a} f(x)\right]^n$

7. $\lim_{x \to a} c = c$

8. $\lim_{x \to a} x = a$

9. $\lim_{x \to a} x^n = a^n$

10. $\lim_{x \to a} \sqrt[n]{x} = \sqrt[n]{a}$

11. $\lim_{x \to a} \sqrt[n]{f(x)} = \sqrt[n]{\lim_{x \to a} f(x)}$

where n is a positive integer.

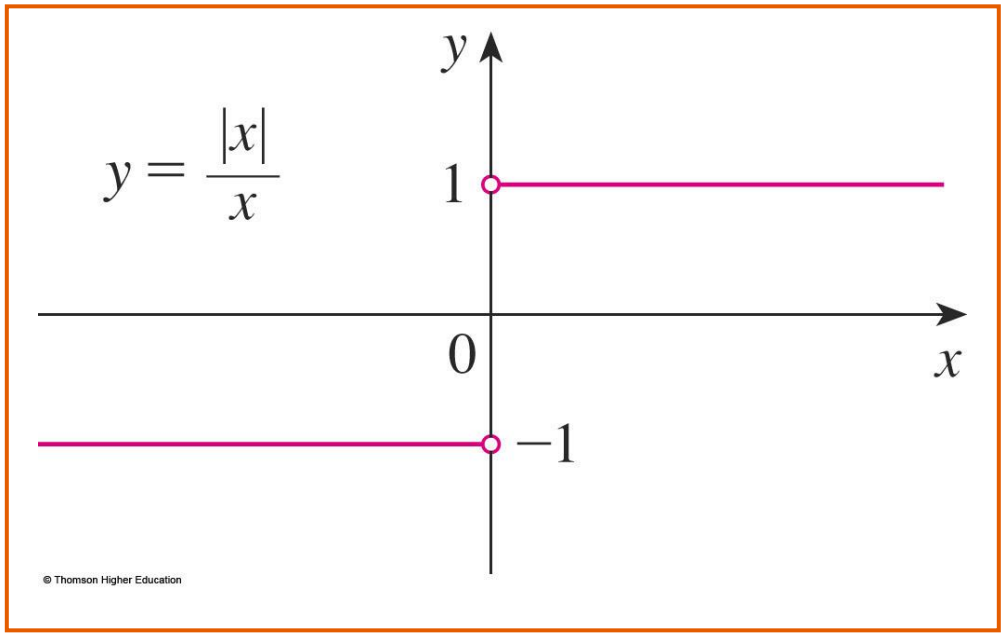
USING THE LIMIT LAWS

$\lim_{x \to a} f(x) = L$ if and only if $\lim_{x \to a^-} f(x) = L = \lim_{x \to a^+} f(x)$

USING THE LIMIT LAWS

Example 4

Prove that $\lim_{x \to 0} \frac{|x|}{x}$ does not exist.



PROPERTIES OF LIMITS

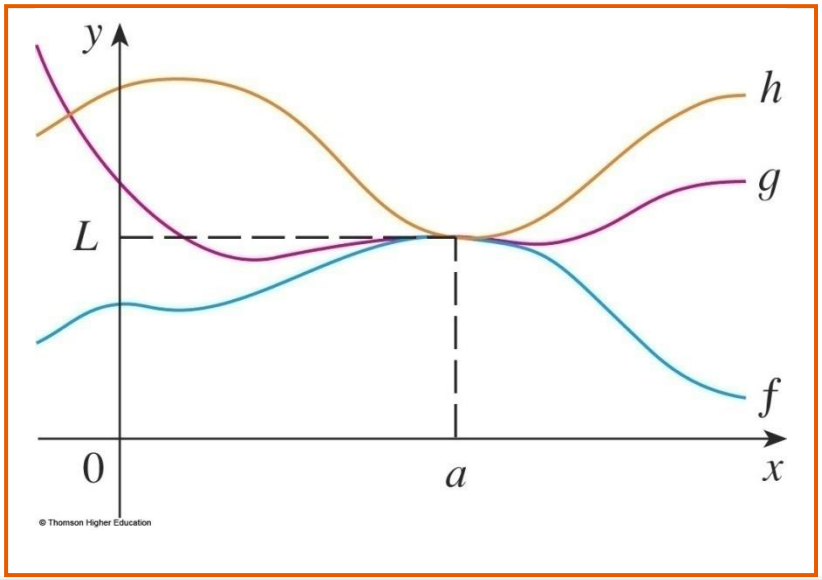
If $f(x) \pounds g(x)$ when x is near a (except possibly at a) and the limits of f and g both exist as x approaches a, then

$$\lim_{x \to a} f(x) \leq \lim_{x \to a} g(x)$$

SQUEEZE THEOREM

The Squeeze Theorem (the Sandwich Theorem or the Pinching Theorem) states that,

if $f(x) \le g(x) \le h(x)$ when x is near a (except possibly at a) and $\lim_{x \to a} f(x) = \lim_{x \to a} h(x) = L$ .Then $\lim_{x \to a} g(x) = L$



USING THE LIMIT LAWS

Example 5

Show that $\lim_{x \to 0} x^2 \sin \frac{1}{x} = 0.$

– Note that we cannot use

$$\lim_{x \to 0} x^2 \sin \frac{1}{x} = \lim_{x \to 0} x^2 \cdot \lim_{x \to 0} \sin \frac{1}{x}$$

– This is because $\lim_{x \to 0} \sin(1/x)$ does not exist.

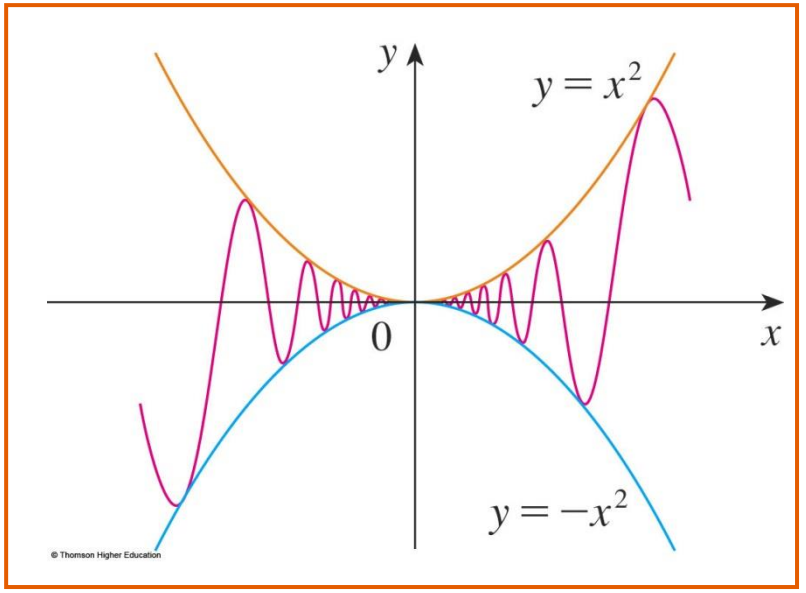
USING THE LIMIT LAWS

Example 5

– However, since $-1 \le \sin\frac{1}{x} \le 1$,

we have: $-x^2 \le x^2 \sin\frac{1}{x} \le x^2$

- Taking $f(x) = -x^2$, and $h(x) = x^2$ in the Squeeze Theorem, we obtain: $\lim_{x \to 0} x^2 \sin\frac{1}{x} = 0$



QUIZ QUESTIONS

1) If $\lim_{x\to 3} f(x) = 0, \lim_{x\to 3} g(x) = 0$

then $\lim_{x\to 3} \frac{f(x)}{g(x)}$ does not exist

a. True

b. False

QUIZ QUESTIONS

If $\lim_{x \to 3} f(x)g(x)$ exists, then the limit must be $f(3)g(3)$

a. True

b. False

LIMITS

2.4 Continuity

In this section, we will:

See that the mathematical definition of continuity corresponds closely with the meaning of the word continuity in everyday language.

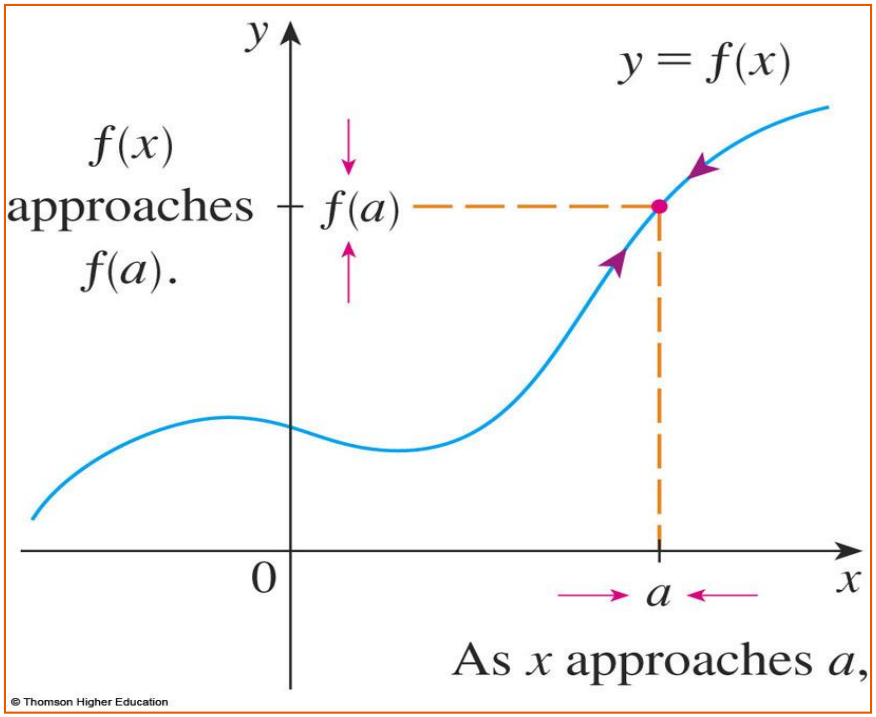
CONTINUITY

A function f is **continuous at a** if:

$$\lim_{x \to a} f(x) = f(a)$$

Notice that :

- $f(a)$ is defined - that is, a is in the domain of f
- $\lim_{x \to a} f(x)$ exists.
- $\lim_{x \to a} f(x) = f(a)$



CONTINUITY

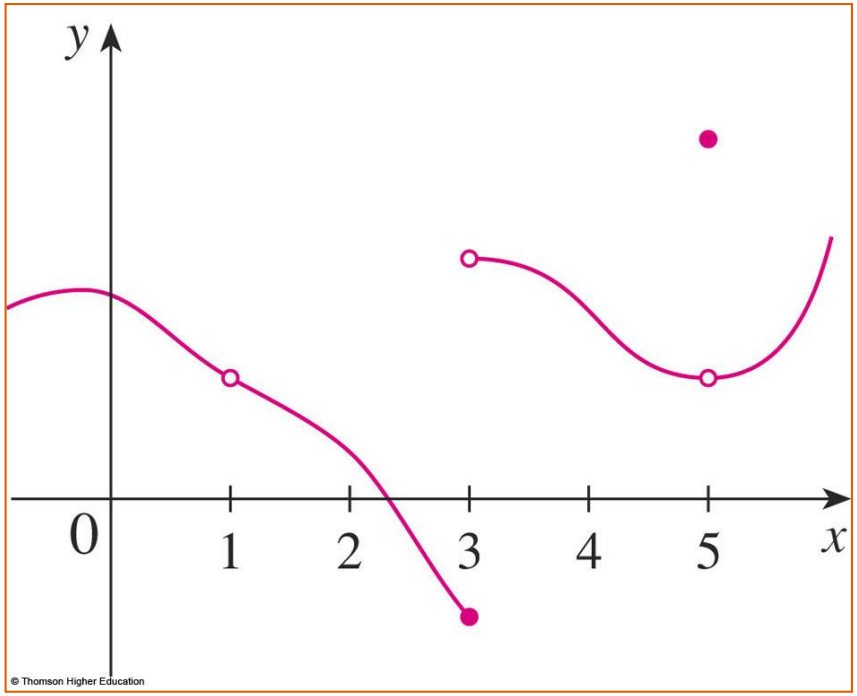
Definition

If f is defined near a - that is, f is defined on an open interval containing a, except perhaps at a - we say that f is discontinuous at a if f is not continuous at a.

The figure shows the graph of a function f.

At which numbers is f discontinuous?

Why?



CONTINUITY

Definition

If $f(x)$ is discontinuous at a, then

1. f has a **removable discontinuity** at a if $\lim_{x \to a} f(x)$ exists. (Note: When we state that $\lim_{x \to a} f(x)$ exists, we mean that $\lim_{x \to a} f(x) = L$, where L is a real number.)
2. f has a **jump discontinuity** at a if $\lim_{x \to a^-} f(x)$ and $\lim_{x \to a^+} f(x)$ both exist, but $\lim_{x \to a^-} f(x) \neq \lim_{x \to a^+} f(x)$. (Note: When we state that $\lim_{x \to a^-} f(x)$ and $\lim_{x \to a^+} f(x)$ both exist, we mean that both are real-valued and that neither take on the values $\pm\infty$.)
3. f has an **infinite discontinuity** at a if $\lim_{x \to a^-} f(x) = \pm\infty$ or $\lim_{x \to a^+} f(x) = \pm\infty$.

Definition

CONTINUITY

A function f is continuous from the right at a number a if

$$\lim_{x \to a^+} f(x) = f(a)$$

and f is continuous from the left at a if

$$\lim_{x \to a^-} f(x) = f(a)$$

CONTINUITY

Definition

A function *f* is continuous on an interval if it is continuous at every number in the interval.

– If *f* is defined only on one side of an endpoint of the interval, we understand ‘continuous at the endpoint’ to mean ‘continuous from the right’ or ‘continuous from the left.’

CONTINUITY

Theorem

If f and g are continuous at a; and c is a constant, then the following functions are also continuous at a:

1. $f + g$
2. $f - g$
3. cf
4. fg
5. $\frac{f}{g}$ *if* $g(a) \neq 0$

CONTINUITY

Theorem

The following types of functions are continuous at every number in their domains:

- Polynomials
- Rational functions
- Root functions
- Trigonometric functions

Theorem

CONTINUITY

If f is continuous at b and $\lim_{x \to a} g(x) = b$ then

$$\lim_{x \to a} f(g(x)) = f(b)$$

In other words,

$$\lim_{x \to a} f(g(x)) = f\left(\lim_{x \to a} g(x)\right)$$

– If x is close to a, then $g(x)$ is close to b; and, since f is continuous at b, if $g(x)$ is close to b, then $f(g(x))$ is close to $f(b)$.

CONTINUITY

Theorem

If g is continuous at a and f is continuous at g(a), then the composite function $(f \circ g)(x) = f(g(x))$ *is continuous at a.*

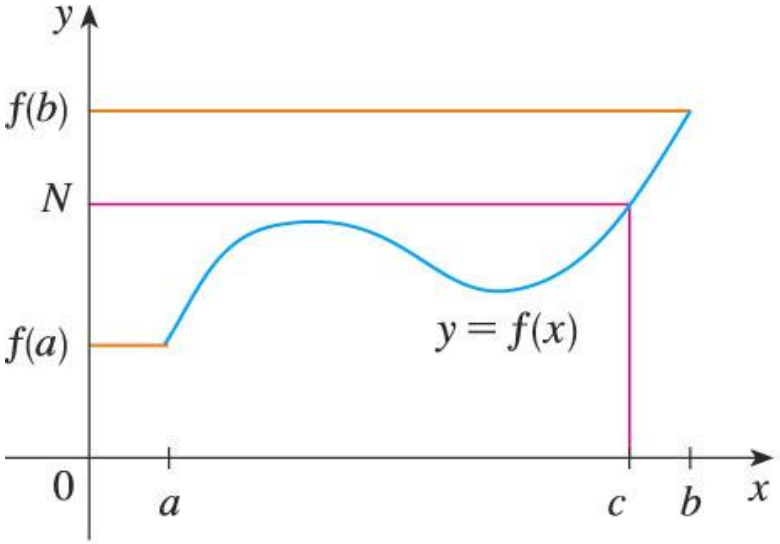
This theorem is often expressed informally by saying "a continuous function of a continuous function is a continuous function."

INTERMEDIATE VALUE THEOREM

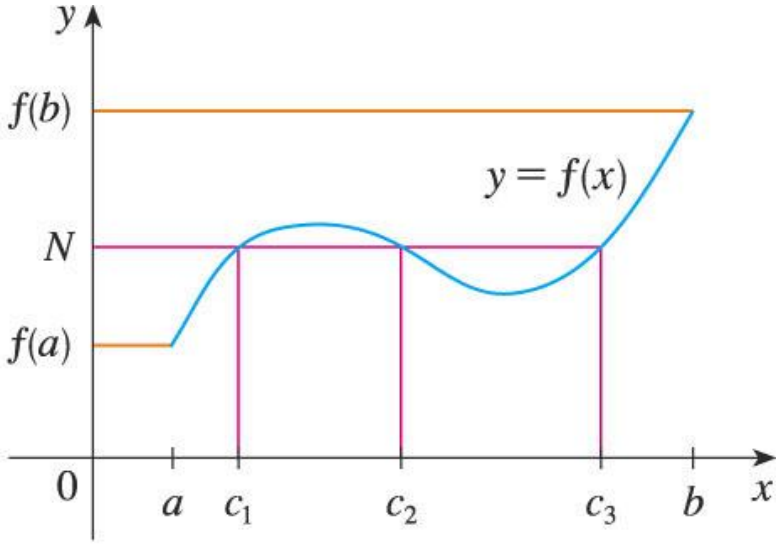
Theorem

Suppose that f is continuous on the ***closed interval [a, b]*** *and let N be any number between f(a) and f(b), where* $f(a) \neq f(b)$

Then, there exists a number c *in* (a, b) *such that* $f(c) = N$.



(a)



(b)

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INTERMEDIATE VALUE THEOREM

Example 6

Show that there is a root of the equation

$$4x^3 - 6x^2 + 3x - 2 = 0$$ between 1 and 2.

– Let $f(x) = 4x^3 - 6x^2 + 3x - .2$

– We are looking for a solution of the given equation

that is, a number c between 1 and 2 such that $f(c) = 0$.

– Therefore, we take $a = 1$, $b = 2$, and $N = 0$ in the theorem.

– We have $f(1) = 4 - 6 + 3 - 2 = -1 < 0$

and $f(2) = 32 - 24 + 6 - 2 = 12 > 0$

QUIZ QUESTIONS

1) If f(1)>0 and f(3)<0 then there exists a number c between 1 and 3 such that f(c)=0

a. False

b. True

2) Which is the equation expressing the fact that " f is continuous at 2"?

a. $\lim_{x\to 2} f(x) = 2$

b. $\lim_{x\to \infty} f(x) = f(2)$

c. $\lim_{x\to 2} f(x) = 0$

d. $\lim_{x\to 2} f(x) = \infty$

e. $\lim_{x\to 2} f(x) = f(2)$

LIMITS AT INFINITY

Let f be a function defined for every x>a. Then

$$\lim_{x\to\infty} f(x) = L \quad \text{means that} \quad \forall \varepsilon > 0, \exists M > 0$$

$$\text{if} \quad x > M \quad \text{then} \quad |f(x) - L| < \varepsilon$$

DEFINITION

The line y=L is called the **horizontal asymptote** of f(x) if we have one of the following:

$$\lim_{x\to\infty} f(x) = L \qquad \lim_{x\to-\infty} f(x) = L$$

Example

Find the asymptotes of the function

$$f(x)=\frac{x^3-1}{x^3+x^2-2}$$

Solution

$$\lim_{x\to\infty}\frac{x^3-1}{x^3+x^2-2}=\lim_{x\to\infty}\frac{1-\frac{1}{x^3}}{1+\frac{1}{x}-\frac{2}{x^3}}=1$$

y=1 is horizontal asymptote

$$\frac{x^3-1}{x^3+x^2-2}=\frac{(x-1)(x^2+x+1)}{(x-1)(x^2+2x+2)}$$

$$\lim_{x\to 1}\frac{x^3+1}{x^3+x^2-2}=\frac{3}{5}$$

Example

Compute

a. $\lim_{x\to\infty} \sin\frac{1}{x}$ — 0

b. $\lim_{x\to\infty} (\sqrt{x^2+1}-x)$ — 0

c. $\lim_{x\to\infty} \sin x$ — Does not exist

d. $\lim_{x\to\infty} (x-x^3)$

QUIZ QUESTIONS

1) Find $\lim_{x \to \infty} \cos x$

a. 0 b. infinity c. 1 d. Does not exist

2) Find $\lim_{x \to \infty} \frac{1}{x} \cos x$

a. 0 b. infinity c. 1 d. Dose not exist

QUIZ QUESTIONS

3) If $\lim_{x\to 0} f(x) = \infty, \lim_{x\to 0} g(x) = \infty$

Then $\lim_{x\to 0}[f(x) - g(x)] = 0$

a. True b. False

4) A function can have two different horizontal asymptotes

a. True b. False